
Light Scattering in Disordered and Nonreciprocal Media

Thesis defense

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Since the foundations of the electromagnetic theory, efforts have been made to understand how light interacts with complex environments.

In this thesis we will address some open questions in magnetoplasmonic systems, on one side, and random correlated systems, on the other.

In the first part, we studied, through simple physical models, the rich phenomenology obtained in hybrid magnetoplasmonic systems. In particular, we determine the exact role of interactions between forming elements of the systems and their contribution to the global magneto-optical response.

In the second part of this thesis we focus our attention on the light emission dynamics (lifetimes or emission rate statistics) in three-dimensional systems with strong structural correlations undergoing structural phase transitions. The objective was to link the statistical properties of emission dynamics with the thermodynamics of the system and prove that there are fundamental differences between lifetime statistics and transport properties.

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